DOUBLE SKIN CHW AIR HANDLING UNIT ENGINEERING DATA SHEET:
1. Thermal break extruded aluminum framework with 50mm injected polyurethane foam (PUF) insulation panels (Density 40 kg/m3).
2. Double skin casing: Outer skin 0.8mm pre-coated GI sheet; Inner skin 0.8mm plain GI with antimicrobial powder coating.
3. Plug fan: Direct-drive backward curved aerofoil impeller dynamically balanced to ISO 1940 Grade G2.5 with IE4 synchronous motor.
4. Cooling coil: 6-row deep copper tube (12.7mm OD) with hydrophilic coated corrugated aluminum fins (12 fins per inch).
5. Dual filtration: Pre-filter MERV 8 (10 Micron, G4) followed by fine secondary bag filter MERV 13 (2.5 Micron, F7).

Parameter	AHU-01 (Chamber)	AHU-02 (Cabinet)	AHU-03 (Lobby)	AHU-04 (Press Hall)
Airflow Capacity	18,000 CFM	12,000 CFM	15,000 CFM	10,000 CFM
External Static Pressure	450 Pa	400 Pa	400 Pa	350 Pa
Cooling Capacity	92.0 TR	61.5 TR	76.8 TR	51.2 TR
Water Flow Rate	220 USGPM	148 USGPM	184 USGPM	123 USGPM
Entering Water Temp	6.7°C (44°F)	6.7°C (44°F)	6.7°C (44°F)	6.7°C (44°F)
Leaving Water Temp	13.3°C (56°F)	13.3°C (56°F)	13.3°C (56°F)	13.3°C (56°F)
Motor Rating	22.0 kW (IE4)	15.0 kW (IE4)	18.5 kW (IE4)	11.0 kW (IE4)

AMARAVATI GOVERNMENT COMPLEX · LEGISLATIVE ASSEMBLY		
DWG TITLE: CHILLED WATER DOUBLE-SKIN AIR HANDLING UNIT T		
DRAWING NO: SL-288 -CHW TYPE AHU-DATA SHEE	REV: A1	SCALE: 1:100
CLIENT: AP CRDA / GOVT OF ANDHRA PRADESH	DATED: 28-09-2026	